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121 Most Asked DSA Interview Questions In Interviews

Master Data Structures & Algorithms

A comprehensive collection of frequently asked questions by FAANG+ companies

Arrays & Strings

1.Two Sum - Find two numbers that add up to a target
2. Best Time to Buy and Sell Stock
3. Contains Duplicate - Check if array has duplicates
4. Product of Array Except Self
5. Maximum Subarray (Kadane's Algorithm)
6. Merge Intervals
7. Rotate Array by k steps
8. 3Sum - Find three numbers that sum to zero
9. Container With Most Water
10. Longest Substring Without Repeating Characters
11. Valid Palindrome
12. Group Anagrams
13. Longest Palindromic Substring
14. Trapping Rain Water
15. Move Zeroes to End
16. Find All Duplicates in Array
17. Missing Number in Array
18. First Missing Positive
19. Spiral Matrix Traversal
20. Set Matrix Zeroes

Linked Lists

21. Reverse Linked List (Iterative & Recursive)
22. Detect Cycle in Linked List
23. Merge Two Sorted Lists
24. Remove Nth Node From End
25. Find Middle of Linked List
26. Palindrome Linked List
27. Add Two Numbers (Linked Lists)
28. Copy List with Random Pointer
29. Flatten a Multilevel Doubly Linked List
30. Intersection of Two Linked Lists
31. Remove Duplicates from Sorted List
32. Reorder List
33. Swap Nodes in Pairs

Stacks & Queues

34. Valid Parentheses
35. Min Stack - Stack with O(1) getMin()
36. Implement Queue using Stacks
37. Daily Temperatures
38. Largest Rectangle in Histogram
39. Sliding Window Maximum
40. Evaluate Reverse Polish Notation
41. Next Greater Element
42. Simplify Path (Unix style)
43. Design Circular Queue

Trees & Binary Search Trees

44. Binary Tree Inorder Traversal
45. Maximum Depth of Binary Tree
46. Validate Binary Search Tree
47. Lowest Common Ancestor of BST
48. Binary Tree Level Order Traversal
49. Invert Binary Tree
50. Diameter of Binary Tree
51. Symmetric Tree
52. Path Sum
53. Serialize and Deserialize Binary Tree
54. Kth Smallest Element in BST
55. Construct Binary Tree from Preorder and Inorder
56. Binary Tree Right Side View
57. Count Good Nodes in Binary Tree
58. Binary Tree Zigzag Level Order
59. Flatten Binary Tree to Linked List
60. Populating Next Right Pointers

Graphs

61. Number of Islands (DFS/BFS)
62. Clone Graph63. Course Schedule (Topological Sort)
64. Pacific Atlantic Water Flow
65. Word Ladder
66. Graph Valid Tree
67. Cheapest Flights Within K Stops
68. Network Delay Time (Dijkstra)
69. Minimum Height Trees
70. Accounts Merge
71. Redundant Connection
72. Critical Connections in Network

Dynamic Programming

73. Climbing Stairs
74. House Robber
75. Coin Change
76. Longest Increasing Subsequence
77. Longest Common Subsequence
78. Word Break
79. Decode Ways
80. Unique Paths
81. Maximum Product Subarray
82. Edit Distance (Levenshtein)
83. Partition Equal Subset Sum
84. Regular Expression Matching
85. Jump Game
86. House Robber II (Circular)
87. Longest Palindromic Subsequence
88. Interleaving String
89. Target Sum
90. Minimum Path Sum

Backtracking & Recursion

91. Permutations
92. Subsets
93. Combination Sum
94. Generate Parentheses95. N-Queens Problem
96. Word Search in 2D Grid
97. Palindrome Partitioning
98. Letter Combinations of Phone Number
99. Sudoku Solver
100. Combinations

Binary Search

101. Binary Search Implementation
102. Find First and Last Position in Sorted Array
103. Search in Rotated Sorted Array
104. Find Minimum in Rotated Sorted Array
105. Search a 2D Matrix
106. Median of Two Sorted Arrays
107. Koko Eating Bananas
108. Time Based Key-Value Store
109. Find Peak Element

Heaps & Priority Queues

110. Kth Largest Element in Array
111. Top K Frequent Elements
112. Merge K Sorted Lists
113. Find Median from Data Stream
114. Task Scheduler
115. Design Twitter (with priority queue)

Bit Manipulation

116. Single Number (XOR)
117. Number of 1 Bits (Hamming Weight)
118. Counting Bits
119. Reverse Bits
120. Missing Number (using XOR)
121. Sum of Two Integers (without + operator)